connected to the pipe. The center of the pipe is 80 mm below the level of mercury. (Sp. Gravity=13.6) in the right limb. If difference of mercury levels in the two limbs is 130mm, find the pressure of oil in the pipe.

Q.29 If $3.7m^3$ of on oil weights 34.25KN, find its specific weight, mass density and relative density.

Q.30 Explain basic components of pneumatic systems.

Q.31 Differentiate between centrifugal and reciprocating pumps?

Q.32 What is the necessity of FRL unit in Pneumatic system.

Q.33 Name various minor losses in pipes.

Q.34 Define the laminar and turbulent flow with example.

Q.35 State the difference between

i) Uniform flow and non-uniform flow

ii) Steady and Unsteady flow.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

Q.36 Write short note on

i) Hydraulic Ram ii) Hydraulic door closer

Q.37 Explain the construction and working of Francis turbine with neat sketch.

Q.38 Explain Bernoulli's theorem. Write its assumptions, limitations and application.

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Roll No. ....

4th Sem / Auto, Mech (3rd/4th), Prod(3rd), T&D(3rd), GE, CNC, Adv, Manuf. Tech., Mechatronics, CAD/CAM, Mech Engg (Fabrication Tech), Mech Engg (CAD/CAM Dsgn & Robotics)

Subject:- Hydraulics and Pneumatics / Hyd. & Hyd. M/c

Time : 3Hrs. M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

Q.1 The fluid which has no viscosity, no surface tension and no compressibility is

a) Ideal fluid b) Real-fluid

c) Newtonian fluid d) Nm-Newtonian fluid

Q.2 The specific weight of water is

a) $1000\ N/m^3$ b) $9810\ N/m^3$

c) $9.81\ N/m^3$ d) $1000\ Kg/m^3$

Q.3 Continuity equation deals with the law of conservation of

a) Mass b) Energy

c) momentum d) none of the above

Q.4 The inlet length of venturimeter is ______ the outlet length.

a) Equal to b) More than

c) Less then d) Any of the above

Q.5 Hydraulic ram works on the principle of
a) Centrifugal b) water hammer
c) Bernoulli's theorem d) Reciprocating action

Q.6 For small discharge at high pressure _____ pump is preferred.
a) Centrifugal pump b) Arial flow pump
c) Propeller pump d) Reciprocating pump

Q.7 The standard value of atmospheric pressure is
a) 760 mm of mercury b) 1.01325 bar
c) 10.34 m d) all of the above

Q.8 When the velocity of flow increases, the pressure
a) Decreases b) Increases
c) Becomes eatable d) none of the above

Q.9 The C_d of an orifice is always
a) greater then c_c b) equal to c_v
c) equal to c_c d) less than c_c

Q.10 Pelton wheel is also called
a) an impulse turbine
b) an impulse-reaction turbine
c) a reaction turbine
d) none of these

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 Weight Density =__________

Q.12 Define surface tension

Q.13 Define manometers

Q.14 Define total Head.

Q.15 What is hydraulic mean depth.

Q.16 Define Nozzle.

Q.17 Capillarity is due to _____ and_______

Q.18 A U-tube manometer is used to measure absolute pressure at any point. (True/False)

Q.19 Venturimeter measures flow of fluids in pipes only when pipe is horizontal.(True/False)

Q.20 Kaplan turbine is a radial flow reaction turbine. (True/False)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Explain Pascal's law and its application.

Q.22 Write short note on Bourdon's Tube pressure gauge.

Q.23 Explain major Head Losses in pipes.

Q.24 A pipe 12.5cm in diameter is used to transport oil of specific gravity 0.75 under a pressure of 1 bar. If the total head is 20m relative to a datum plan which is 2.5m below the center of the pipe. Find its discharge.

Q.25 Write short note on hydraulic jack.

Q.26 Differentiate between real fluid and ideal fluids with suitable examples.

Q.27 Define the following terms.
i) Specific volume ii) Capillarity
iii) Surface tension

Q.28 A simple manometer is used to measure the pressure of oil (sp. Gravity=0.9) flowing in a pipeline. Its right limb is open to the atmosphere and left limb is